

NeuroView

20 lessons in team science: Learning from the experience of the International Brain Laboratory

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Leveraging large-scale neuroscience datasets requires new collaborative approaches. The International Brain Laboratory (IBL) is a distributed, open-science experiment facing these challenges. To succeed, IBL developed methods for shared decision-making, division of labor, authorship/credit assignment, career support, standardization, and robust data analysis. We share these lessons learned to benefit other collaborative scientific efforts.

Introduction

In the last decade, neuroscience has experienced a dramatic shift toward large-scale data acquisition, open-science practices of sharing and reusing data and code, and team-based science where a large number of individual contributors each bring deep expertise in their domain. These changes reflect a maturing scientific field where the expense and infrastructure requirements of data acquisition, curation, exploration, and dissemination are growing beyond the capabilities of typical individual laboratories. In parallel, there is a growing recognition that the benefits of consortia outweigh the organizational costs. While there remains an important role for research in individual labs, it is now common for neuroscientists to participate in massive multi-year team efforts to gather data (e.g., HCP,¹ mouse CCF,² BICCN,³ and MICrONS⁴) and develop data acquisition technologies and standards (e.g., Neuropixels,⁵ NWB,⁶ and BIDS⁷), and these are growing in scale and scope every year.

In line with these efforts, the International Brain Laboratory (IBL) is an experi-

ment in pursuing large-scale reproducible science in a distributed and collaborative environment. The IBL was founded with the intention to “develop a new way of doing neuroscience in partnership across multiple laboratories, sharing experimental protocols, data, and analyses to ensure tight collaboration and high reproducibility.”⁸ Our consortium has largely succeeded in this goal with both large-scale projects with reproducibility at their core and dozens of **smaller individual projects**.

Here, we collect 20 lessons we have learned in creating and executing the vision of the IBL that we hope other consortia may find applicable to their own efforts.

Working together

The IBL modeled itself on the principles of sociocracy using a consent-based approach (Figure 1A) that brought numerous benefits: we kept decisions small enough and sufficiently deliberative so that every stakeholder could agree that a decision was worth trying, without endangering the organization. We found

that **(1) collaborative governance requires consistent and balanced participation to achieve success**; while only a handful of members had the time or interest to fully participate, we were still able to effectively generate and revise critical governance policies. We also found that **(2) maintaining a continuous process of coming to consensus and reviewing policies mitigated conflict**. For example, the IBL publication policy received five significant revisions during the publication process of dozens of papers, with the most recent revision in 2024. Many reasonable disagreements arise when comparing traditional “first author” approaches with IBL’s attempts to acknowledge the collaborative efforts of the entire consortium. A consensus-based decision process provides a framework for adapting policies to reflect hard-won experience.

Any member of the IBL can contribute a proposal and participate in policy discussions, but final votes are the responsibility of a body known as the general assembly (Figure 1B). In response to concerns about lack of representation, the general

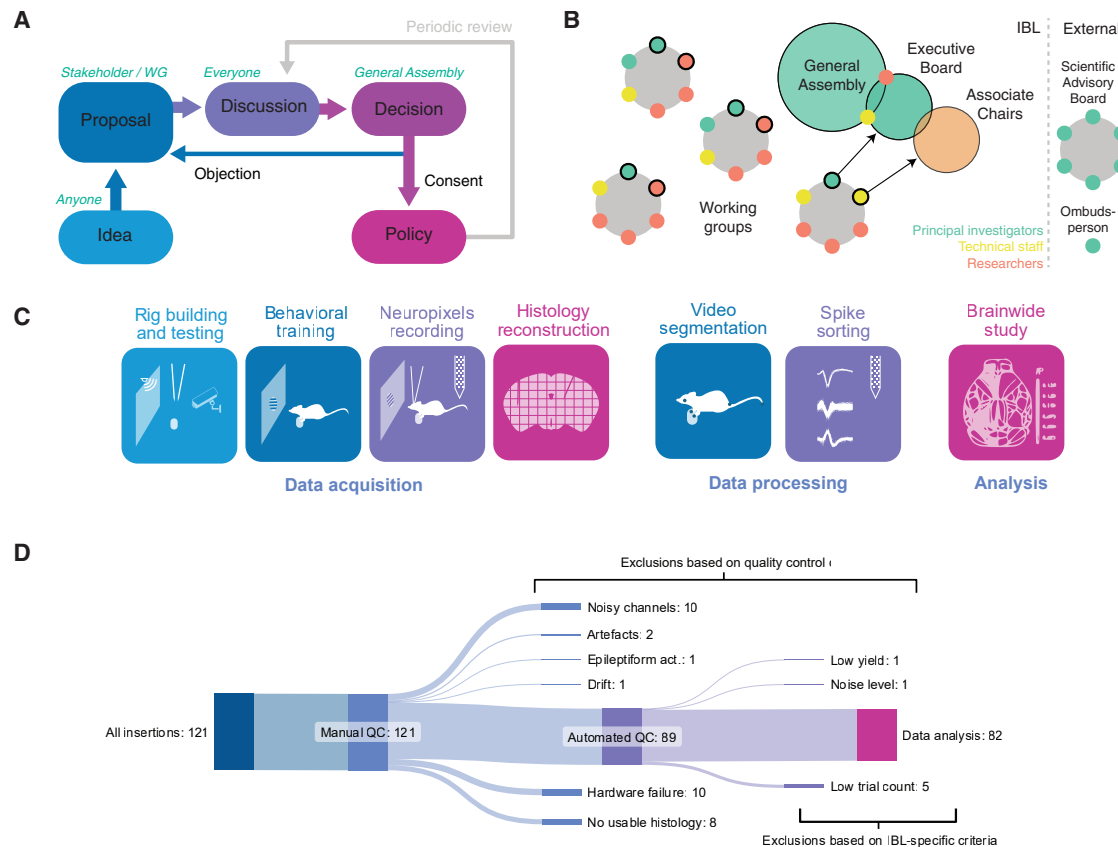


Figure 1. Consensus, organization, and rigorous quality control: Three pillars of the IBL

(A) The International Brain Laboratory (IBL) consensus-based decision process involves a group drafting a new policy proposal, circulating it for feedback, and finally coming to consensus within the general assembly for ratification. Review dates ensure continuous evaluation, amendment, and renewal of the IBL policies.

(B) The IBL is organized as a flattened hierarchy inspired by ATLAS at CERN. Working groups defined around data acquisition, analysis, and large milestones are led jointly by a chair (principal investigator [PI]) and associate chair (researcher or staff). These leaders create the executive board and associate chair group, respectively, and the full group of PIs along with rotating researcher and staff representatives form the general assembly. PI effort is typically a small fraction of their time, while researcher involvement varies across projects and over time. An external scientific advisory board and ombudsperson provide additional guidance and support. (Figure adapted from Lauren E. Wool and International Brain Laboratory.⁵)

(C) Reproducibility is driven by standardization and quality control at every stage of the project life cycle, from data acquisition and processing to analysis.

(D) Achieving reproducibility requires many checks at each stage and often requires rejecting many data assets. Examples of quality control checks in the IBL include validation of rig hardware, behavioral training criteria, checks on raw Neuropixels data, and metrics calculated during histology alignment, pose estimation, and curation of units.

assembly was expanded from only principal investigators to include a representative for researchers (for research assistants, graduate students, and postdocs) and staff. This enlarged membership of the general assembly was ratified in the IBL charter and reflects our recognition that **(3) decision-making bodies should include representatives from every level of the organization**. But representation does not ensure a voice for all members, and **(4) multiple strategies are necessary to include a broad range of perspectives in a large organization**. To gather regular anonymous feedback, we implemented an annual survey. Detailed responses are first reviewed by

the researcher and staff representatives and then compiled by a dedicated working group. Results are then openly discussed among all members during annual meetings. For issues that require immediate turnaround, a separate anonymous feedback form is available at any time of year, though it has been used very rarely. Although we have worked to create an environment with balanced participation, the organization remains situated within the wider academic landscape; as in most research settings, principal investigators ultimately manage resources and guide trainees' career paths, which can make trainees reluctant to raise concerns.

From early in the IBL's existence, we recognized that **(5) credit assignment in a large consortium requires a different approach**. Concerns about authorship are even more challenging in a large collaboration, because contributions often fit outside of traditional categories. We ultimately settled on two very different solutions. First, the IBL authorship policy specifies that large platform papers will include all members whose contributions have not yet been described in another publication. For example, large-scale projects rely on a wide range of contributions, from data acquisition by technicians to contributions to software and hardware engineering, and the IBL

recognizes these critical contributions through authorship on platform papers. The second approach is that junior researchers are encouraged to leverage the IBL platform (both existing data and infrastructure) to develop their own smaller-scale projects to answer specific questions. Together, these approaches balance the importance of highlighting all contributions while recognizing that **(6) project leadership remains critical to successful career development in academia, even as team science grows in importance**. IBL researchers have taken advantage of this opportunity, and more than 20 research projects are currently being supported (a few examples can be found on [our website](#)). The authorship policy also provides a standard framework for generating authorship lists and for using CRediT¹⁰ to self-identify the contributions of each author. A standardized CRediT contributions matrix is incorporated into each of our papers, including in [Figure S1](#) of this paper. For consortia in neuroscience to flourish, the community must recognize a wide range of contributions to scientific research and find creative ways to ensure that individual leaders stand out.

The IBL brought together theorists and experimentalists to develop the best possible experimental design informed by theoretical predictions. However, in order to address concerns about the practicality of replicating a single experiment in many labs, the final design was shaped as much by experimental constraints as theoretical ideas. As data collection began, theorists were recruited to work on processing and data analysis, taking resources away from theoretical explorations. Theoretical approaches finally became central when the dataset was complete, with several new theory papers released in 2024 and 2025. Many further insights will come from the cross-analysis of the IBL surveys with other large- and small-scale datasets. Our experience is that **(7) continual integration of theory and experiment at scale is challenging**, and most of the smaller-scale projects at the IBL have involved only a single lab, either experimental or theoretical. Between-lab scientific exchange requires focused, dedicated effort on the part of everybody involved and is difficult to sustain at a distance.

As the IBL has matured, we have increasingly recognized that **(8) researchers and research software engineers (RSEs) are both critical to large-scale, reproducible research**. RSEs have been instrumental to the success of the IBL, developing and maintaining an open-access codebase supporting rig control, automated processing, and comprehensive outward-facing documentation alongside spearheading the development of several neuroscience software packages ([Pinpoint](#), [Lightning Pose](#), [alignment GUI](#), [IBL sort](#)) actively used by the wider community. However, overreliance on a few key RSEs can become a bottleneck, especially when turnover occurs. Building resilience in an organization requires researchers and engineers to work together—ideally with both groups engaged in data scrutiny, tool development, bug reporting, and documentation, ensuring shared responsibility and continuity of institutional knowledge. At the IBL, we now also **(9) recognize RSEs as a distinct scientific role that is essential to sustaining infrastructure for scientific discovery**. Formal, supported career paths that reward engineering contributions are required to reliably build and maintain the robust, reusable open-source tools that large-scale neuroscience demands. As an example, the [Lightning Pose](#) software used to analyze behavioral videos was developed by a joint graduate student and postdoc team within one of our academic labs. The IBL infrastructure enabled the postdoc to transition to a research software engineer position with responsibility for maintaining and advancing the tool, supporting long-term adaptability and growth.

While RSE career growth is now supported by several scientific organizations, it is less widely acknowledged that **(10) researchers and technical staff have different priorities for the speed of scientific discovery versus robustness and reusability**. Many of the most persistent problems in collaborative science within the IBL (data quality, software maintenance, and onboarding to complex systems) were ultimately solved by IBL technical staff. Academic incentives do not reward the foundational work to develop platforms, and while PhD students and postdocs invested substantial

time in developing infrastructure and toward data acquisition, their time was still limited by pressure to work on traditional smaller-scale projects in parallel. To address this, we are increasingly supportive of “industrialized” workflows in which dedicated, technical, and highly trained individuals perform the bulk of the data collection and infrastructure work (e.g., [Allen Institute for Neural Dynamics OpenScope program](#)). This model relieves pressure on early-career researchers while ensuring consistency and quality.

Standardization and building for scale

Research software comes with a set of very different requirements compared to traditional labs, and in our experience, **(11) achieving reproducibility across multiple laboratories requires extensive standardization with shared experimental protocols and automated data pipelines**. Standardized approaches also enable economies of scale: tools, workflows, and infrastructure can be maintained more efficiently than a proliferation of bespoke pipelines. The IBL’s modular infrastructure allows experimental components to be swapped, extended, or removed to meet changing researcher needs. However, **(12) in large-scale neuroscience, even minor errors can propagate and become costly**. During rig assembly early in the IBL, a researcher accidentally reversed the two wires of a rotary encoder with no immediate visible consequences. The resulting polarity error silently corrupted behavioral data for months, affecting downstream analyses across multiple labs. Although the technical fix was straightforward and quality assurance checks ensure that the error cannot re-occur, it still took months to audit affected sessions, communicate the scope of the problem, and coordinate re-running key analyses. This leads to our next lesson: **(13) large-scale data collection introduces a tension between consistency and correction**. Once thousands of sessions have been acquired, uncovering an error raises difficult decisions: should flawed protocols be preserved for consistency or revised, risking dataset fragmentation? Early decisions become costly to reverse. When we discovered a flaw in how camera timestamps were acquired,

each lab had to modify their hardware setup, resulting in delays to data collection. For prior datasets, significant engineering effort was required to retrospectively recover timestamps. At scale, even small adjustments carry large implications and make adaptability a strategic necessity.

An increasing number of papers in neuroscience rely on the output of complex pipelines for the processing of behavior, spike sorting, calcium imaging, and other modalities. Our experience in the IBL has been that **(14) the field needs to invest far more in processing pipelines** and **(15) even widely used software packages need testing before scaling**. For example, IBL engineers identified a long-standing bug in a popular spike-sorting algorithm designed for Neuropixels probes: the algorithm dropped 7 ms out of every 2 s of data (see [GitHub issue](#)). Such discoveries underscore that even foundational analytical tools require ongoing scrutiny and maintenance. Similarly, we weren't able to scale existing pose estimation tools despite standardizing video acquisition across over a dozen rigs and generating extensive training datasets with thousands of labels. These limitations motivated the development of new, more robust pose estimation software. Together, these examples highlight that tools that are considered core infrastructure demand continuous validation, well-curated benchmark datasets, comprehensive test coverage, and iterative improvement.

Early missteps at the IBL also exposed for us the risks of favoring personal preference over technical rigor when designing infrastructure. Although a reliable setup for behavioral control and data acquisition already existed (using MATLAB and NI-DAQ), the IBL initially leaned into an open-source solution in the spirit of open science. However, this choice led to serious issues: logging high-frequency rotary encoder data to text files without sufficient tests in place later revealed dropped lines and inconsistent signals. Engineers were forced to identify and correct corrupted recordings, requiring substantial time and effort. We now **(16) treat infrastructure as a professional engineering project: we take the time to pilot, test continuously, and always visualize outputs immediately**. These improvements have been essential for building a platform

that remains robust and sustainable as the collaboration continues to grow.

In individual labs, quality control can sometimes be an afterthought with a mix of manual checks and existing tools. At the IBL, we found that **(17) end-to-end quality control is necessary to ensure reliable, high-impact neuroscience data and requires sustained effort and coordination**. We implemented comprehensive validation at every step of the experimental pipeline to catch potential issues ([Figures 1C and 1D](#)). Staff and researchers invested significant time in building and iterating on tools, interpreting flagged issues, and tracing anomalies back to root causes. Starting downstream analysis early and getting researchers involved ensured that critical issues were flagged promptly. Automated quality assurance both before and after data acquisition helped minimize misconfiguration of hardware and prevented flawed sessions from progressing downstream. However, we found that **(18) manual review and visualization are still crucial for accurate quality control**. Subtle artifacts can be difficult to define as metrics in advance. For example, we continue to encounter cases where spike-detection algorithms fail to capture the true underlying signals: missing spikes, detecting the same spike multiple times, or mistaking noise for neural activity. Frequent and systematic visualization of both raw and processed data, often overlapped, ensures that these kinds of subtle unknown unknowns are discoverable to researchers and staff. Ultimately, robust quality control processes are what lead to reproducibility and bringing quality control forward in our processes ensures that fewer experiments are rejected during data analysis ([Figure 1D](#)).

The IBL embraces open sharing as a core principle, enabling other scientists to access both data and tools freely. Open access accelerates progress by fostering transparency, inviting external validation, and allowing the community to build rapidly upon our work. However, **(19) while open sharing accelerates innovation, it requires significant investment, and (20) making data useful to the community requires deliberate choices about format, infrastructure, and long-term responsibility**. Data availability is more than just uploading

files: interoperability is imperfect, storage and bandwidth are expensive, and researchers must weigh priorities when choosing formats that trade off between speed, standardization, and machine readability. At the IBL, our data platform is optimized for high-performance querying across tens of thousands of experiments using a relational backend. This environment is tailored to the needs of live, collaborative workflows and supports rapid iteration. In contrast, formats like Neurodata Without Borders⁶ are designed for immutable archives, making them ideal for finalized datasets. No single data-sharing method is optimal for all use cases; trade-offs must be made with downstream users in mind, and multiple formats may be necessary.

Lessons learned: The future of the IBL

The IBL is an ambitious experiment in large-scale neuroscience collaboration, highlighting both the promise and challenges of team-based science. Perhaps the key challenge we have faced over the past decade has been finding a balance across scales: supporting the goals of the collaboration (open science, reproducibility, scale, and longevity) while also supporting the goals of individuals (career advancement, efficiency, and shorter tenure).

As the IBL moves beyond our current accomplishments into a new phase, we plan to share our core resources directly with the neuroscience community. We will transition from a single, large-scale collaboration to a **central core staff** that can support multiple projects with a range of scientific partners. In this way, we hope to enable new teams to perform otherwise impossible collaborative projects. The status quo of bespoke individual labs targeting single brain regions with ad hoc analysis approaches and limited-to-non-existent data sharing is unsustainable for neuroscience. Our aim is to leverage the lessons we have learned and enable routine collaborative and reproducible science at a brain-wide scale.

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AUTHOR CONTRIBUTIONS

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DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.neuron.2026.01.012>.

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